EURUS ELECTRICAL TEAM

RECRUITMENT TASKS

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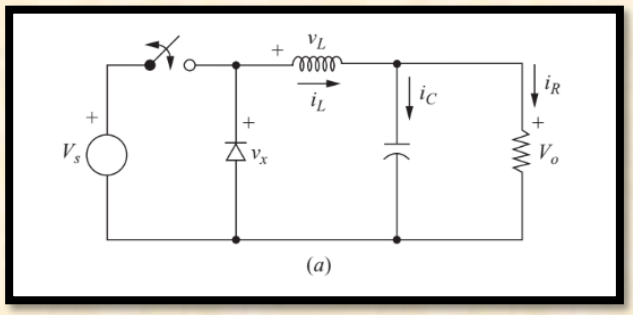
Design of a Buck Converter:

A buck converter, also known as a step-down converter, is a type of DC-DC power converter that takes an input voltage and reduces (or steps down) it to a lower output voltage. The basic operation of a buck converter involves the use of a switch, an inductor, a diode, and a capacitor.



Buck converters are widely used in various electronic devices, such as voltage regulators in power supplies, battery chargers, and energy-efficient LED drivers. They play a crucial role in modern power electronics by efficiently managing and controlling electrical power.

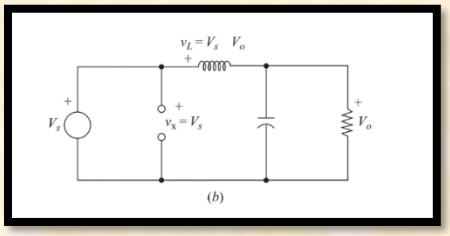
When switch is closed:

$$v_L = V_s - V_o = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L}$$

$$\frac{di_L}{dt} = \frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s - V_o}{L}$$

$$(\Delta i_L)_{\text{closed}} = \left(\frac{V_s - V_o}{L}\right)DT$$ ----------> Eqn. (1)



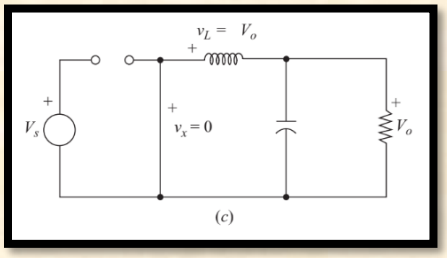
When switch is opened:

$$v_L = -V_o = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{-V_o}{L}$$

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = -\frac{V_o}{L}$$

$$(\Delta i_L)_{\text{open}} = -\left(\frac{V_o}{L}\right)(1-D)T$$ ----------> Eqn. (2)



The output:

The net change in inductor current over one period is zero.

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\left(\frac{V_s - V_o}{L}\right)(DT) - \left(\frac{V_o}{L}\right)(1-D)T = 0$$

$$V_o = V_s D$$ ----------> Eqn. (3)

Inductor current ripples:

From Eqn. (1) and Eqn. (3) we got,

$$\Delta i_L = \left(\frac{V_s - V_o}{L}\right)DT = \left(\frac{V_s - V_o}{Lf}\right)D = \frac{V_o(1-D)}{Lf}$$

So, the value of inductance for a specified peak-to-peak inductor current for continuous-current operation:

$$L = \left(\frac{V_s - V_o}{\Delta i_L f}\right)D = \frac{V_o(1-D)}{\Delta i_L f}$$

Output Voltage Ripple:

While the capacitor current is positive, the capacitor is charging. From the definition of capacitance,

$$Q = CV_o$$

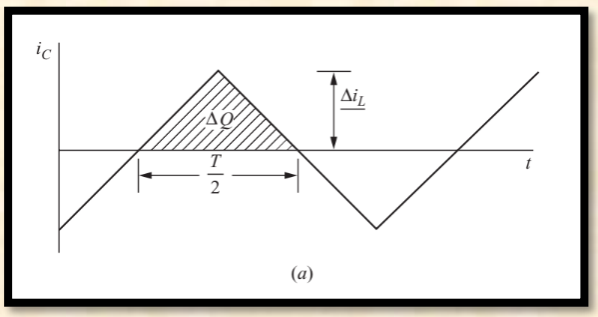
$$\Delta Q = C\Delta V_o$$

$$\Delta V_o = \frac{\Delta Q}{C}$$

The change in charge Q is the area of the triangle above the time axis

$$\Delta Q = \frac{1}{2}\left(\frac{T}{2}\right)\left(\frac{\Delta i_L}{2}\right) = \frac{T\Delta i_L}{8}$$

$$\Delta V_o = \frac{T\Delta i_L}{8C}$$



From Eqn. (2) we got,

$$\Delta V_o = \frac{TV_o}{8CL}(1-D)T = \frac{V_o(1-D)}{8LCf^2}$$

$$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2}$$

capacitance in terms of specified voltage ripple:

$$C = \frac{1-D}{8L(\Delta V_o/V_o)f^2}$$

References:

D. W. Hart, “Power Electronics,” McGraw-Hill Companies Inc., New York, 2010.

Consider a Buck converter circuit with the following specifications:

- Rated input voltage 24 V
- Rated power 100 W
- Rated output voltage 12 V
- Inductor current ripples of 1A
- Output voltage ripples of 5%

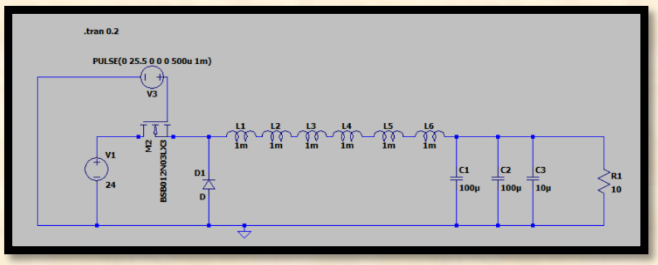
V_{in} = 24 V V_o = 12 V P_w = 100W ΔI_L = 1 A $\frac{\Delta V_o}{\Delta V} = 0.05$ f = 1000Hz T = 1ms

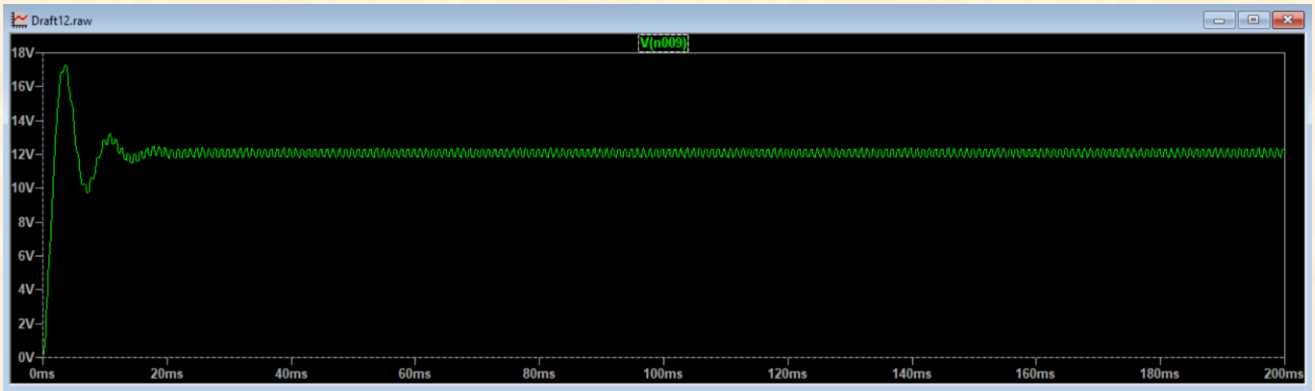
$$D = \frac{V_o}{V_{in}} = 0.5$$

$$D = \frac{T_{on}}{T} \qquad \therefore T_{on} = D * T = 0.5 * 1ms = 500\mu s$$

$$L = \frac{V_o\,(1-D)}{\Delta I_L * f} = \frac{12*0.5}{1*1000} = 6mH$$

$$C = \frac{1-D}{8*L*\frac{\Delta V_o}{\Delta V}*f^2} = \frac{0.5}{8*6mH*0.05*1000^2} = 208\mu F$$





Component/Store	ram electronic	future electronics
Inductor 1mH	Price = 0.75 EGP https://ram-e-shop.com/product/l-0-25w-1mh/	
Capacitor 100uF	Price = 2.25 EGP https://ram-e-shop.com/product/c-100u100v/	
Mosfet	Price = 35 EGP https://ram-e-shop.com/product/irf1010/	Price = 18 EGP https://store.fut-electronics.com/collections/power-mosfet/products/irf1010n-mosfet-55v-85a
Diode	Price = 0.5 EGP https://ram-e-shop.com/product/1n4007/	Price = 1 EGP https://store.fut-electronics.com/collections/diodes/products/diode-1n4001-1a-50v

Generator Type Selection:

DC Generator:

Working Principle: DC generators operate on the principle of electromagnetic induction. The stator contains the field windings, and the rotor (armature) consists of conductors. As the wind turbine rotates the rotor, a magnetic field is created, inducing a direct current in the armature windings.

Advantages:
-Simple design.
-Efficient for small-scale applications.
-Well-suited for battery charging systems.

Disadvantages:
-Requires regular maintenance due to commutators and brushes.
-Limited scalability for higher power applications.
-More expensive for larger capacities.

AC Synchronous Generator:

Working Principle: AC synchronous generators have a rotor and a stator. The rotor's magnetic field synchronizes with the frequency of the grid, inducing an alternating current in the stator windings. They often use permanent magnets or a separate DC power source for field excitation.

Advantages:
-Efficient and reliable.
-Well-suited for grid-connected systems.
-Can handle variable wind conditions effectively.

Disadvantages:
-Complex control systems for synchronization.
-Higher initial cost compared to some other options.

AC Asynchronous Generator:

Working Principle: AC asynchronous generators operate on electromagnetic induction. The rotor induces a current in the stator windings, creating an alternating current. They are often used with a power electronics system to convert the generated AC to DC for grid connection.

Advantages:
-Simple and rugged design.
-Cost-effective.
-Requires less maintenance compared to DC generators.

Disadvantages:
-Lower efficiency compared to synchronous generators.
-Needs power electronics for grid connection.

Choice for a 1 kW Turbine:

For a small wind turbine with a rated power of 1 kW, the choice between these generators depends on various factors. Considering the size and simplicity, a DC generator might be suitable, especially if the application involves battery charging or small off-grid power needs. However, AC asynchronous generators are also a viable option due to their cost-effectiveness and relatively lower maintenance. The final decision should consider factors like the specific requirements of the project, budget constraints, and the ease of integration into the intended application.

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Wind turbine generator types and design to harness the wind. (n.d.). Alternative Energy Tutorials. https://www.alternative-energy-tutorials.com/wind-energy/wind-turbine-generator.html

Chakrabarti, S. (2023, August 3). Types of Wind Turbine Generators and their Functions. SolarFeeds Magazine. https://www.solarfeeds.com/mag/types-of-wind-turbine-generators-and-their-functions/#1_DC_Generator

Cao, W., Xie, Y., & Zheng, T. (2012). Wind Turbine Generator Technologies. In InTech eBooks. https://doi.org/10.5772/51780

MPPT Control:

MPPT: MPPT (Maximum Power Point Tracking) is a technique used to optimize the power output of wind turbines by tracking the maximum power point of the turbine. This is achieved by adjusting the electrical load on the turbine to match the optimal operating conditions for the given wind speed. There are several methods to apply MPPT for converters of wind turbines, but two common methods are Perturb and Observe (P&O) and Incremental Conductance (IC).

P&O: The P&O method involves perturbing the operating point of the turbine and observing the resulting change in power output. The direction of the perturbation is determined by comparing the power output before and after the perturbation. If the power output increases, the perturbation is continued in the same direction until the power output starts to decrease. At this point, the direction of the perturbation is reversed, and the process is repeated until the maximum power point is reached.

IC: The IC method, on the other hand, uses the incremental conductance of the turbine to determine the optimal operating point. The incremental conductance is the change in power output with respect to the change in voltage or current. By monitoring the incremental conductance, the controller can determine the direction of the optimal operating point and adjust the electrical load accordingly.

For small wind turbines: the P&O method may be more suitable due to its simplicity and lower computational requirements. However, the choice of method ultimately depends on the specific requirements and constraints of the wind turbine system.

References:

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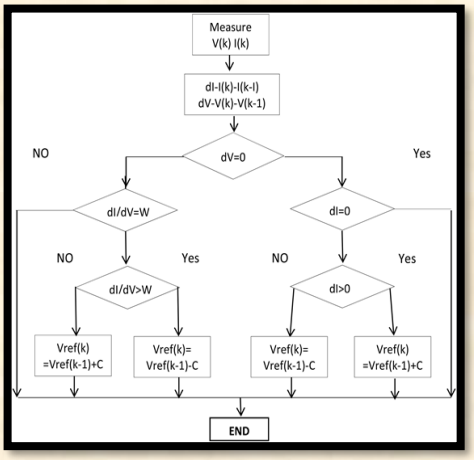
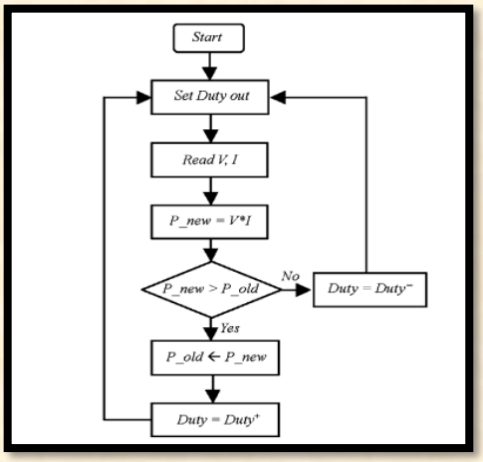
Ramadan, H., Youssef, A., Mousa, H. H. H., & Mohamed, E. E. M. (2019). An efficient variable-step P&O maximum power point tracking technique for grid-connected wind energy conversion system. SN Applied Sciences, 1(12). https://doi.org/10.1007/s42452-019-1716-5

Yu, K. N., & Liao, C. K. (2015). Applying novel fractional order incremental conductance algorithm to design and study the maximum power tracking of small wind power systems. Journal of Applied Research and Technology, 13(2), 238–244. https://doi.org/10.1016/j.jart.2015.06.002



IC Flow Chart



P&O Flow Chart

Motor Speed Controller Design: